Local opportunity does not ensure end-to-end gain

Observed case	Favorable local measurement	Realized outcome
M4 · HumanEval30 Preregistered, fixed k = 2	$V_2^{\text{micro}} = 1.022–1.076$ Shared Metal path; separate micro study	→ ● **Exact + profitable** 1.387× · 90/90 exact
M4 · representative 24Q Frozen representative study	$V_2^{\text{micro}} = 1.022–1.076$ Shared Metal path; separate micro study	→ ● **Exact + negative** 0.842× · 72/72 exact
RTX 5070 Ti Laptop CUDA batched verification	$V_2^{\text{micro}} = 0.983–1.009$ Near one target-decode step	→ □ **Correctness blocked** No admissible speedup
358H · NPU draft Qualified Phase-3 path	48.08 vs 30.81 tokens/s NPU vs CPU · isolated, 8,192 tokens	→ ● **Exact + negative** 0.352× · 90/90 exact

Categorical cases, not a hardware ranking. Filled = qualified; open = blocked. Details and CIs: Figures 2–4.

Micro cost ≠ complete target work; isolated throughput ≠ stateful cost. Speedups cover post-prefill decode.

Execution path

M4 Metal
prefix 512 / 2,048 / 8,192

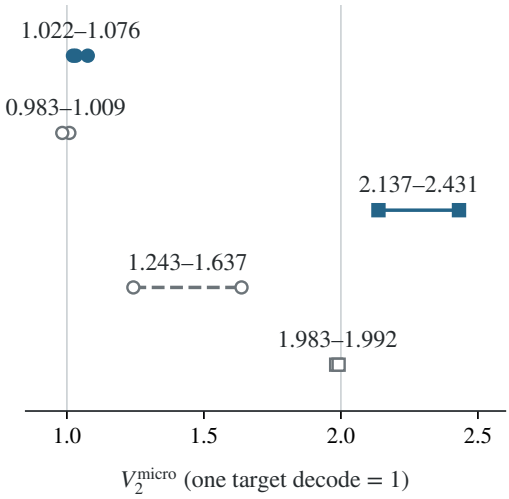
RTX 5070 Ti Laptop · batched
CUDA · ubatch = 2 · depth 0 / 256

RTX 5070 Ti Laptop · serialized
CUDA · ubatch = 1 · depth 0 / 256

B570 Vulkan · fast
Vulkan · context 512 / 2,048

B570 Vulkan · serialized
Vulkan · ubatch = 1 · 512 / 2,048

Measured target-forward cost



Correctness qualification

Qualified
Exact in tested cells

Blocked
Frozen workload gate failed

Qualified control
Exact secondary control

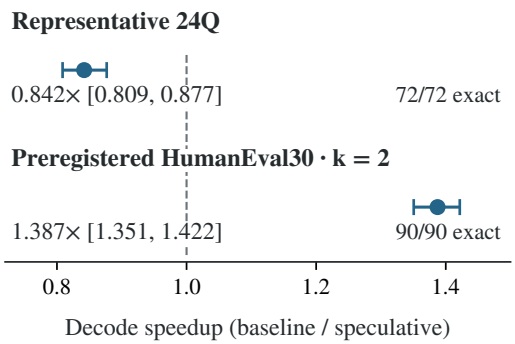
Diagnostic · blocked
45/90 pairs exact

Diagnostic · blocked
60/90 exact; still inadmissible

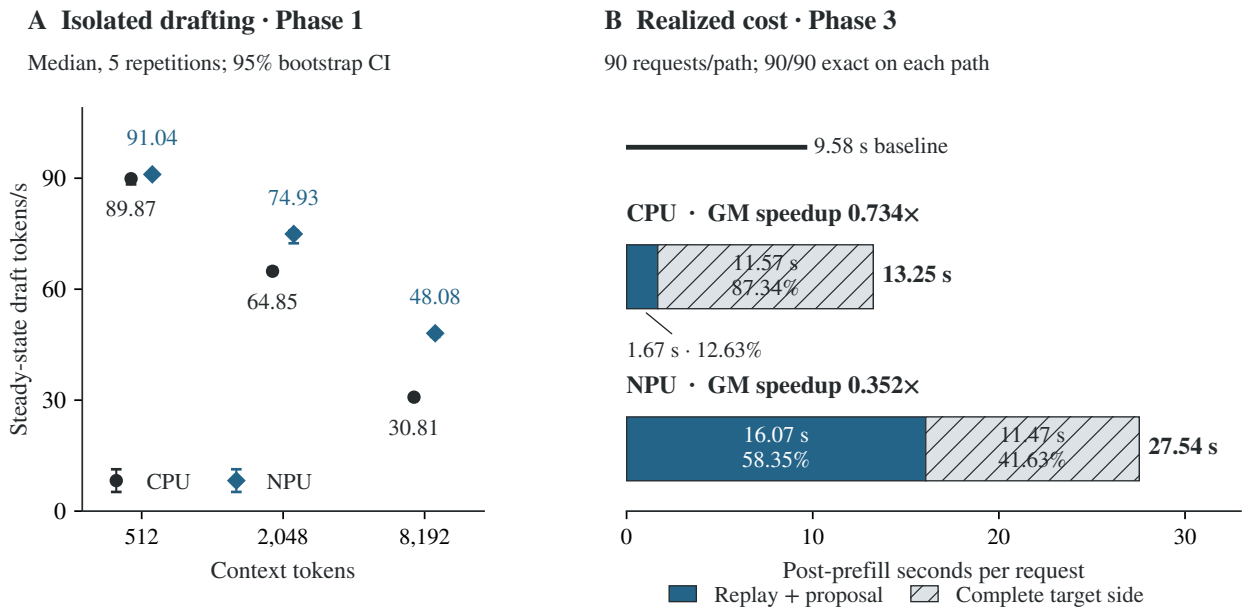
Filled = qualified; open = blocked/diagnostic. Circles = batched/fast; squares = serialized.
Ranges span tested contexts (not CIs); contexts are not fully matched. $V_2^{\text{micro}} \neq R_{\text{target}}$.

Native MLSys width proof: single 3.25 in / double 6.75 in. Print at 100%.

M4 Metal: two frozen workload studies



95% prompt-bootstrap CIs; separate workload studies.



Residual: CPU 4.81 ms (0.04%); NPU 5.06 ms (0.02%). Bars: arithmetic means; speedups: prompt GMs.
Replay and proposal share one timed call. Phase-1 steady-state throughput excludes the full-prefix state cost.
Speedup 95% CIs: CPU [0.693, 0.771]; NPU [0.333, 0.369]. Initial prompt prefill is excluded from Phase 3.

Native MLSys width proof: single 3.25 in / double 6.75 in. Print at 100%.

Categorical regimes under the frozen correctness contract

Mechanism	Not admissible	Correctness-qualified	
	No admissible speed claim	Negative ($S < 1$)	Profitable ($S > 1$)
Workload conditioning		● **M4 representative 24Q [P]** 72/72 exact	● **M4 HumanEval30 [P]** 90/90 exact
Verification constraint	○ **RTX batched [M]** Cheap micro path; gate fails	● **RTX serialized [S]** CPU→RTX / RTX→RTX; exact	
Correctness barrier	○ B570 fast: 45/90 [D] ○ B570 serialized: 60/90 [D]		
Target-side cost		● **358H CPU [P/M]** 90/90; target side 87.34%	
State-interface cost		● **358H NPU [P/M]** 90/90; replay + proposal 58.35%	

P primary · S secondary · M mechanism · D diagnostic. Fractions count exact token-ID pairs.